

Physiology: Lung Mechanics – Elastic Properties of the Lungs

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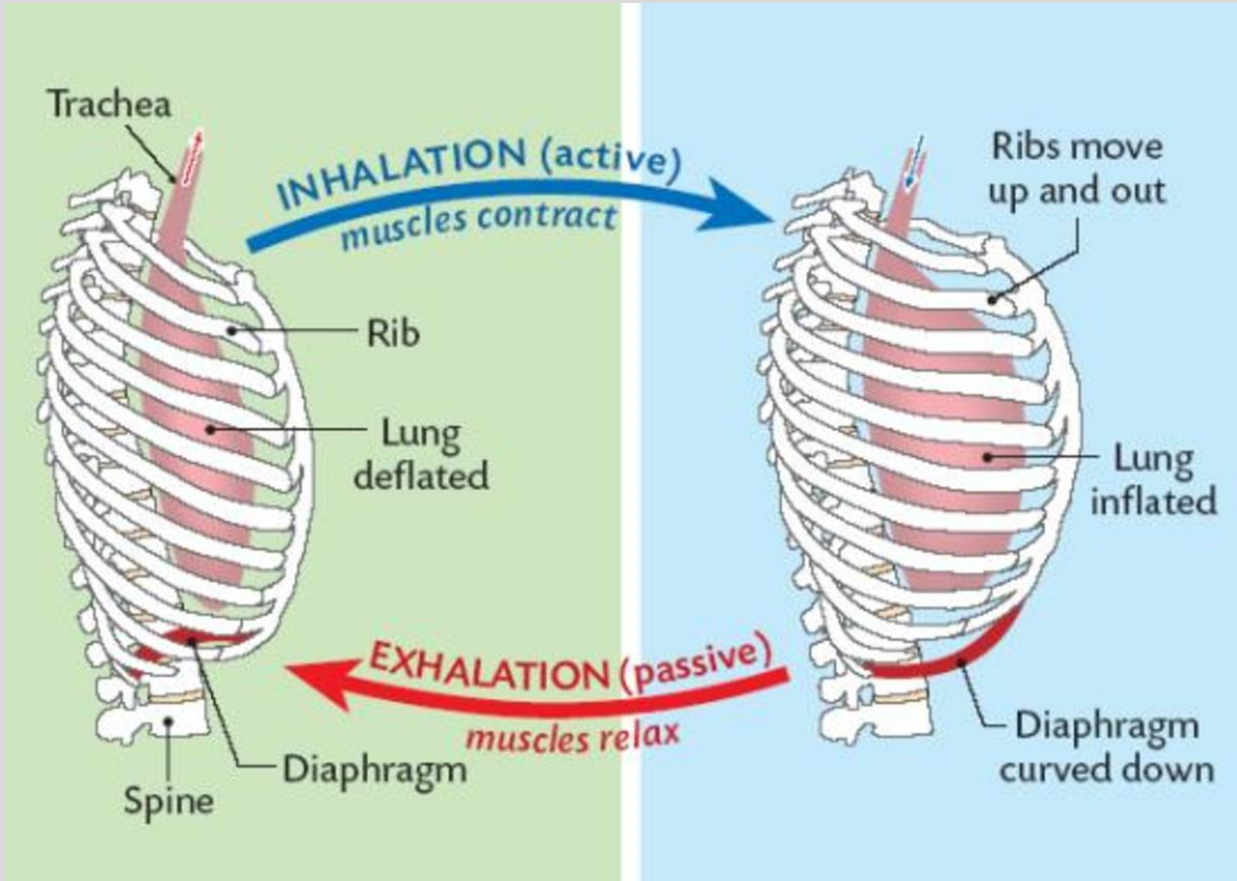
Do.
Make.
Innovate.
Reinvent the Future.

Objectives

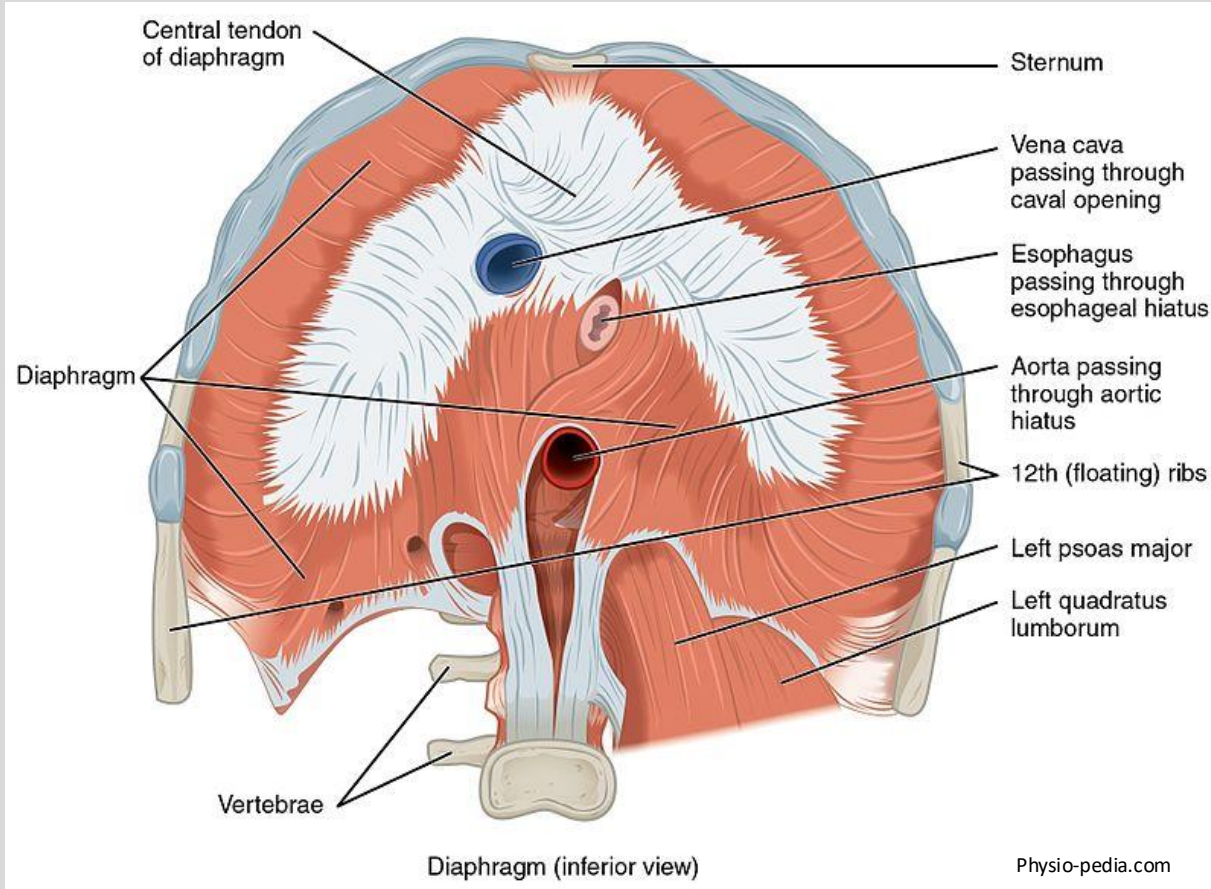
- Name the muscles of respiration and assign their contributions to quiet and active breathing.
- Explain how two pleural layers and intrapleural fluid contribute to transpulmonary (transmural) pressure.
- Explain how compliance and surface tension influence the pressure-volume loop of respiration.
- Compare and contrast the various forces acting on the lungs-chest wall system and how each of them influences transpulmonary pressure. Explain how these forces are altered in individuals with emphysema and fibrosis.
- Explain how the Laplace Law determines surface tension of alveoli and how it factors into neonatal respiratory distress syndrome.
- Describe air flow, pressure, and resistance relationships and common factors that can alter airway resistance.
- Compare and contrast forced expiration of a healthy individual with one suffering from chronic obstructive pulmonary disease (COPD) such as emphysema.

Quiet Breathing

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Muscles of Respiration - Diaphragm



Muscles of Respiration

Muscles of inspiration

Accessory

Sternocleidomastoid
(elevates sternum)

Scalenes Group
(elevate upper ribs)

Not shown:
Pectoralis minor

Principal

External intercostals
Interchondral part of
internal intercostals
(also elevates ribs)

Diaphragm
(dome descends, thus
increasing vertical
dimension of thoracic
cavity; also elevates
lower ribs)

Muscles of expiration

Quiet breathing

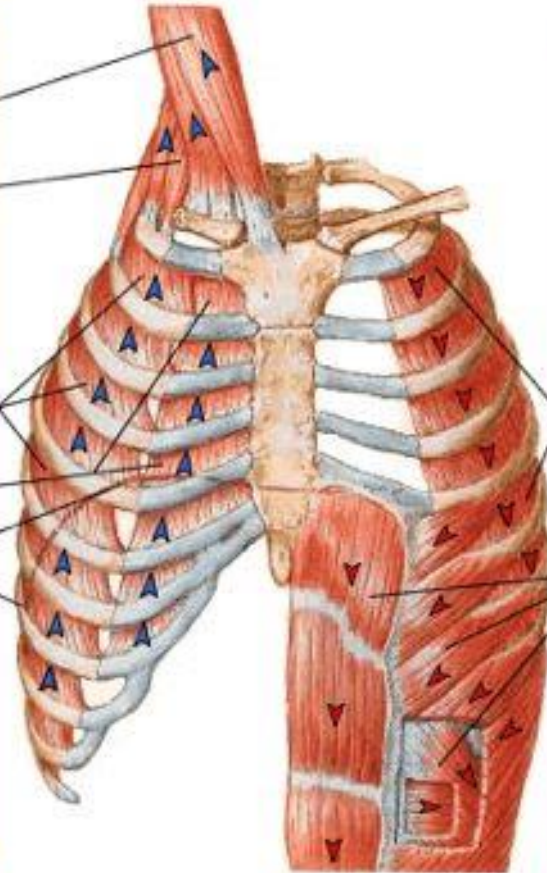
Expiration results from
passive, elastic recoil
of the lungs, rib cage
and diaphragm

Active breathing

Internal intercostals,
except interchondral
part (pull ribs down)

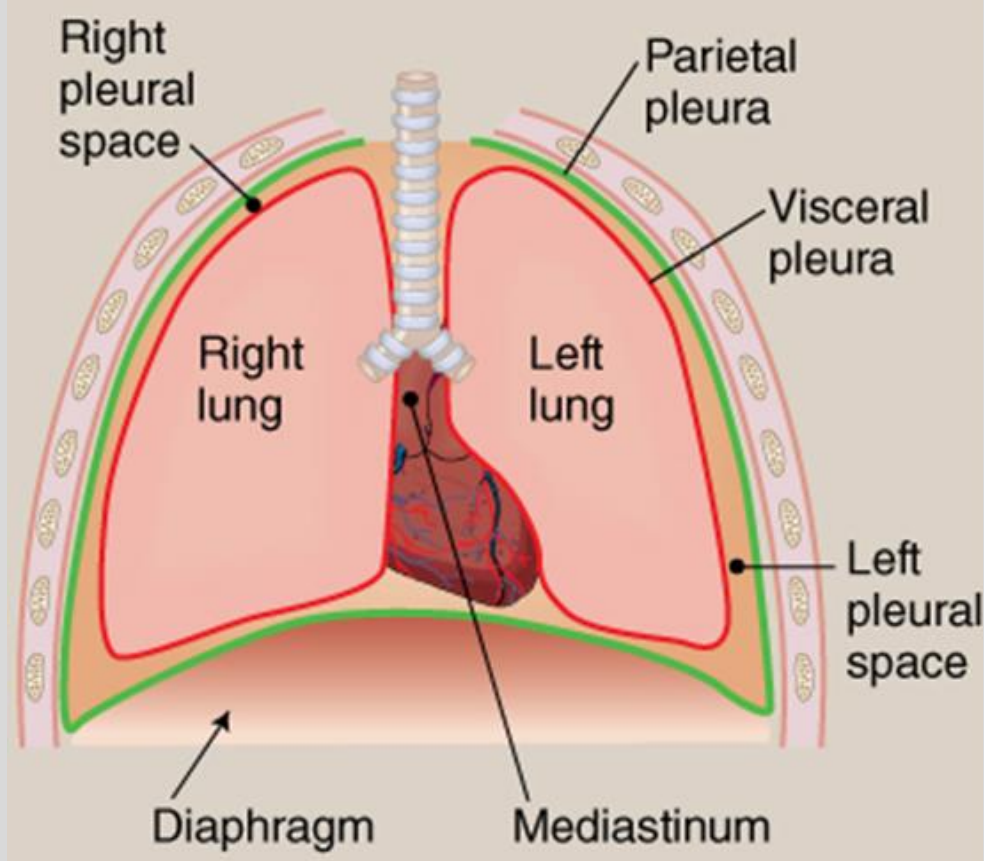
Abdominals
(pull ribs down,
compress abdominal
contents thus pushing
diaphragm up)

Note shown:
Quadratus lumborum
(pulls ribs down)



Used during exercise,
or diseases
with increased
airway resistance

Pleurae



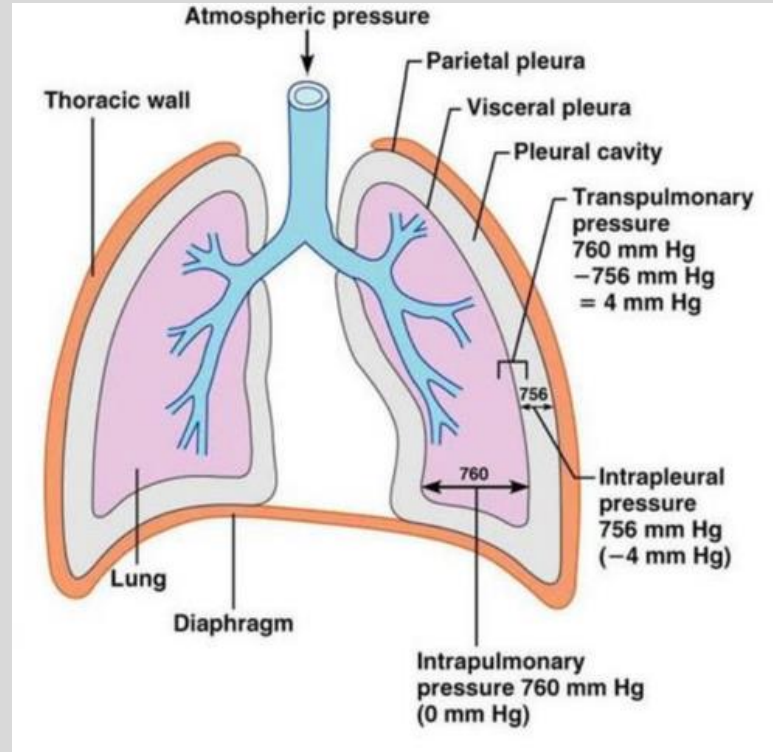
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The chest wall, diaphragm, and mediastinum (heart, larger blood vessels, airways, and associated structures) are covered with **parietal pleura**

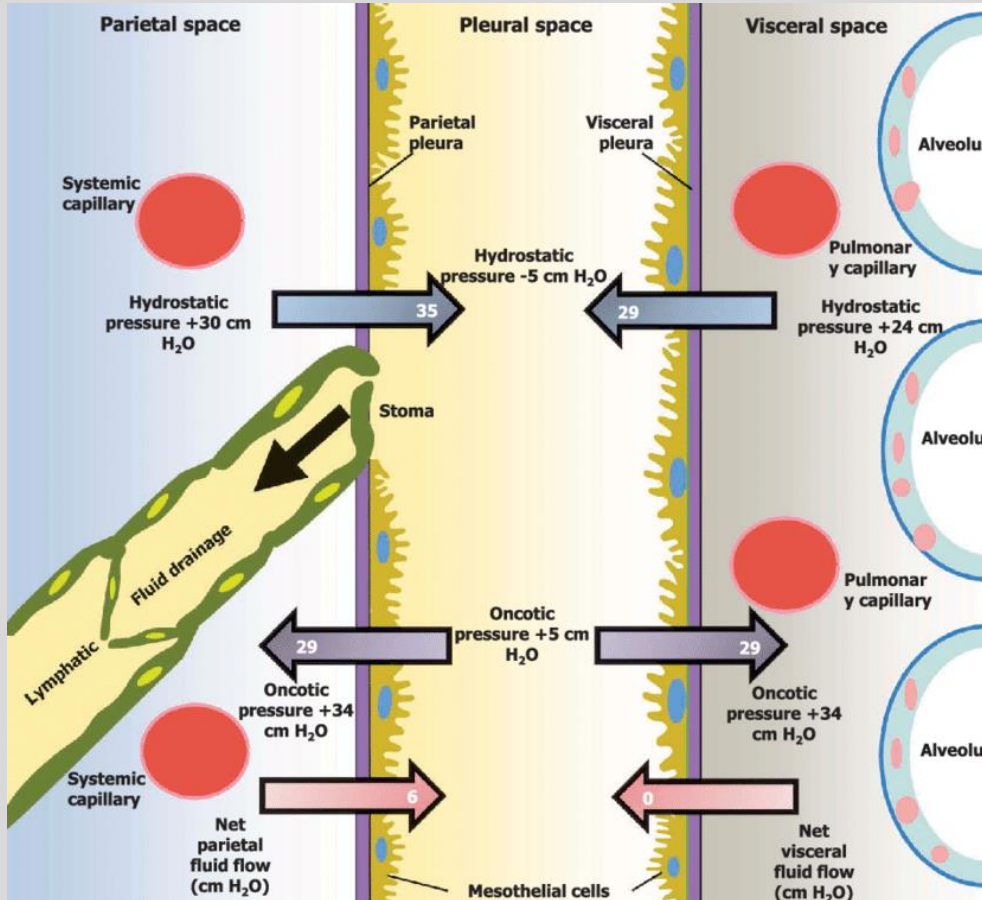
Also: 'Interpleural Space'

Terminology

- Lung pressures referred to atmospheric pressure and 0 when equal to it
- Positive Pressure > 0
- Negative Pressure < 0
- Transmural Pressure:
Pressure inside vs. out
- Transpulmonary Pressure:
Pressure between alveolar space & pleural cavity



Pleural Fluid – Cohesion & Lubrication



About 10ml of total volume

Compliance vs. Elastance (Elasticity)

Change of Volume for a Given Pressure

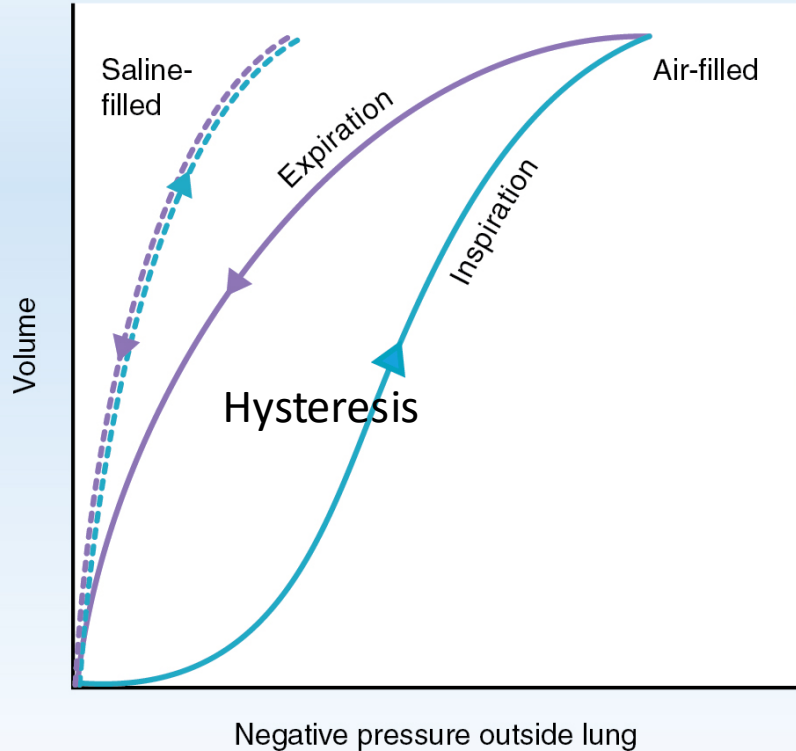
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Lung Compliance: Pressure Volume Loop

Compliance vs. Surface Tension

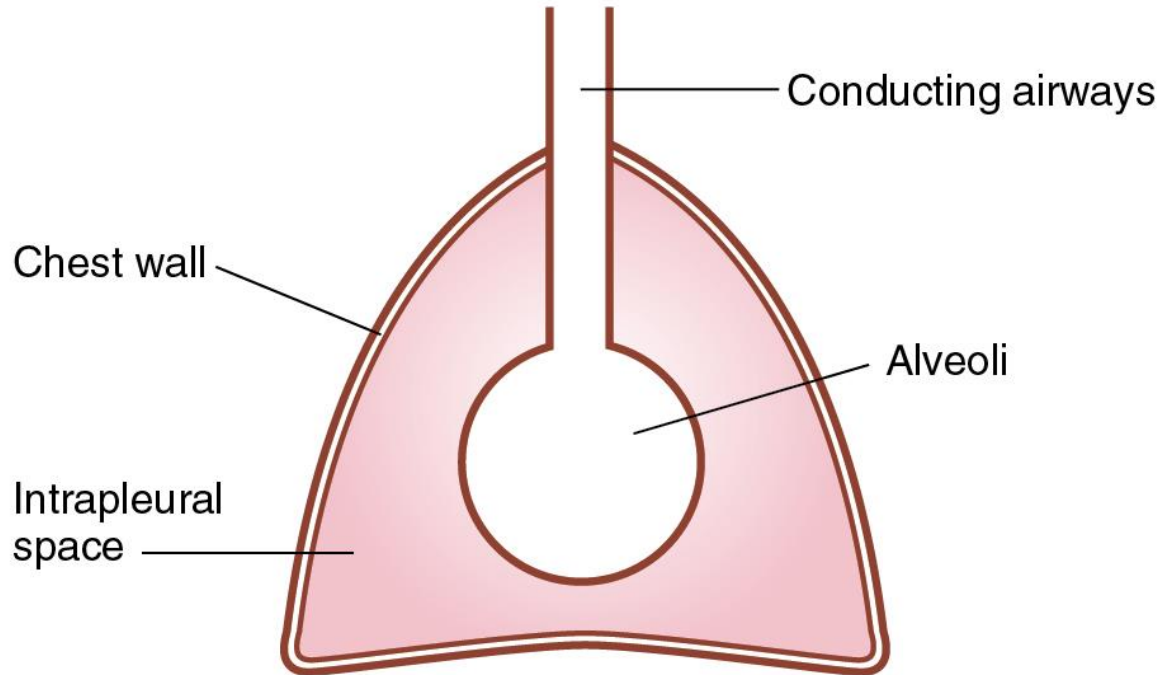
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Compliance Inspiration < Compliance Expiration
Why?

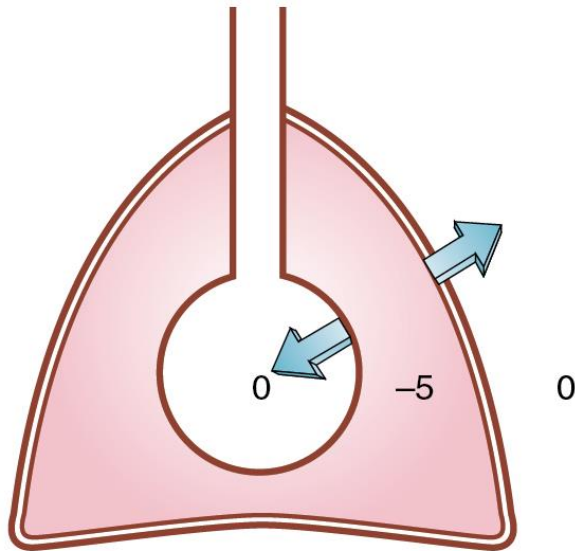
Chest Wall Compliance

LUNG AND CHEST WALL

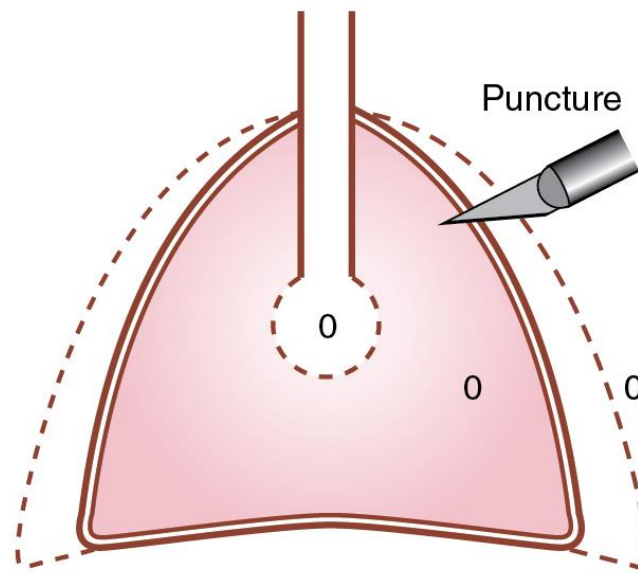


Lungs & Chest Wall: Two Opposing Pressures

Normal

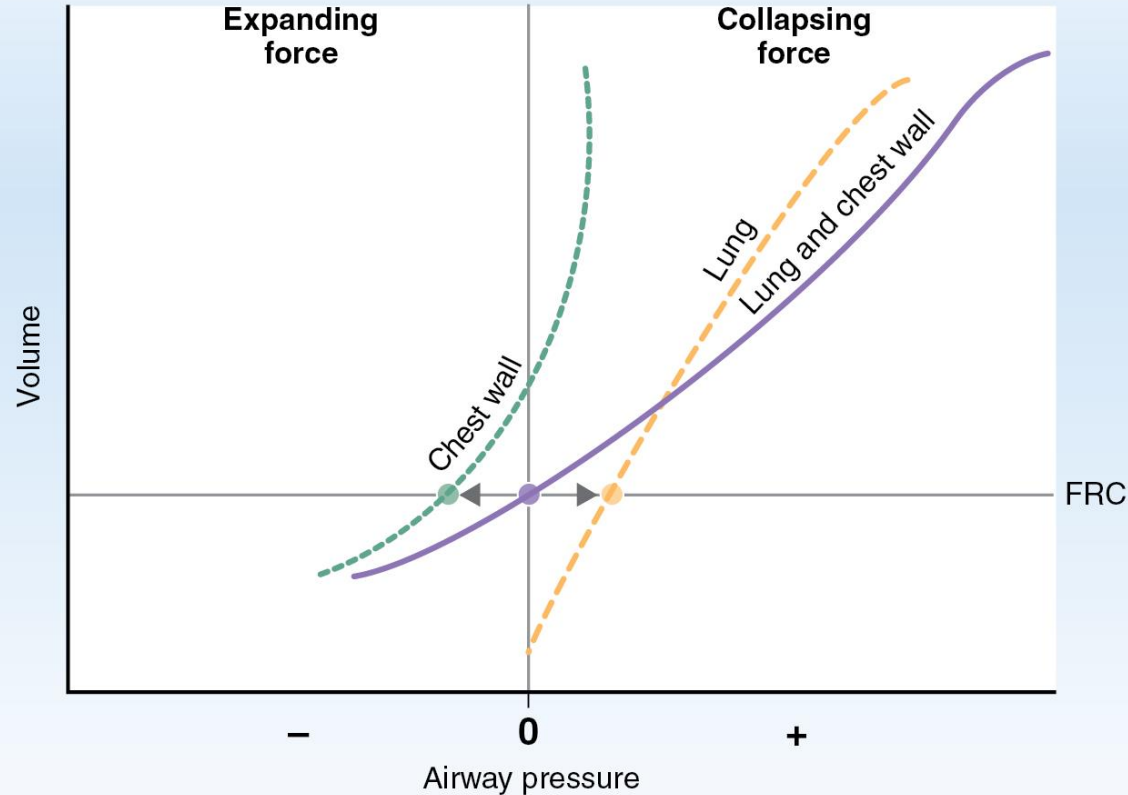


Pneumothorax



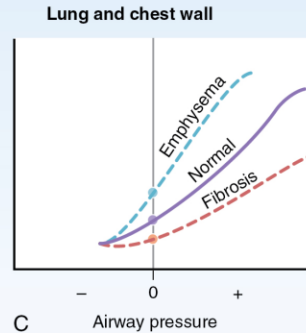
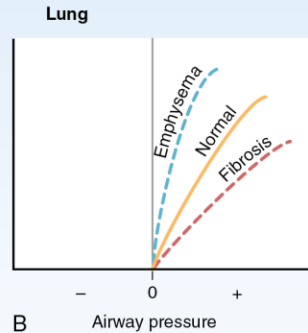
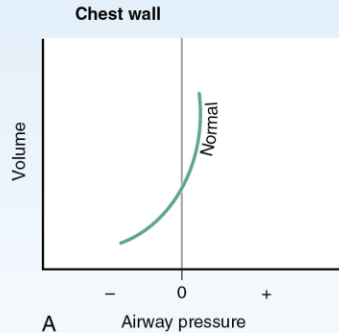
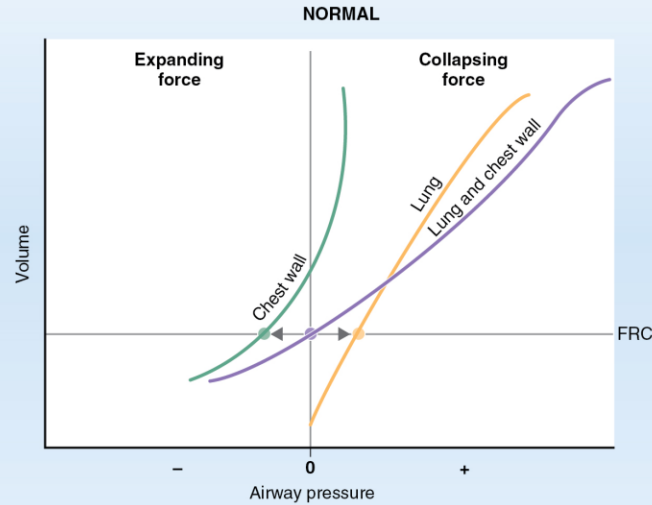
Numbers: cm of H₂O

Compliance of the Lungs-Chest Wall System



Functional Residual
Capacity (FRC)

Diseases Affecting Lung Compliance



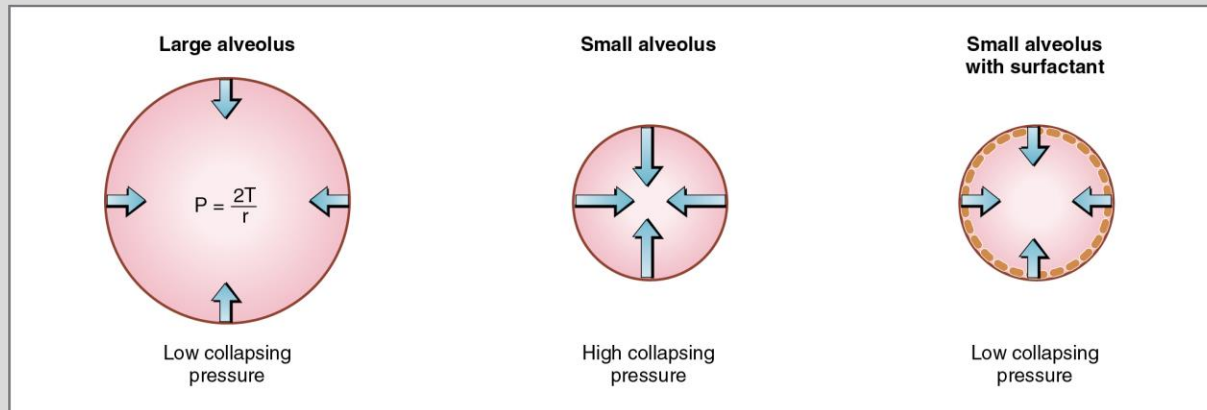
Higher FRC
Normal FRC
Lower FRC

Surface Tension of Alveoli

The Law of Laplace (small spheres)

$$P = \frac{2T}{r}$$

- P = Collapsing pressure on alveolus(dynes/cm²), **or** pressure required to keep alveolus open (dynes/cm²)
- T =Surface tension (dynes/cm)
- r =Radius of the alveolus (cm)

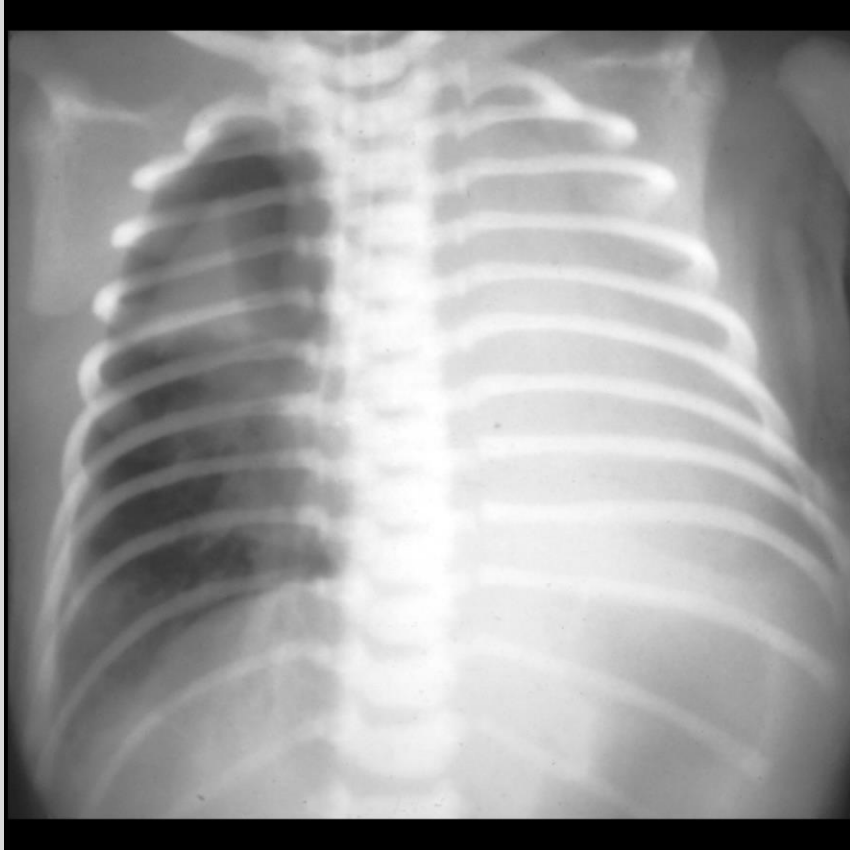


Increases lung compliance
Prevents atelectasis
Ensures uniformity

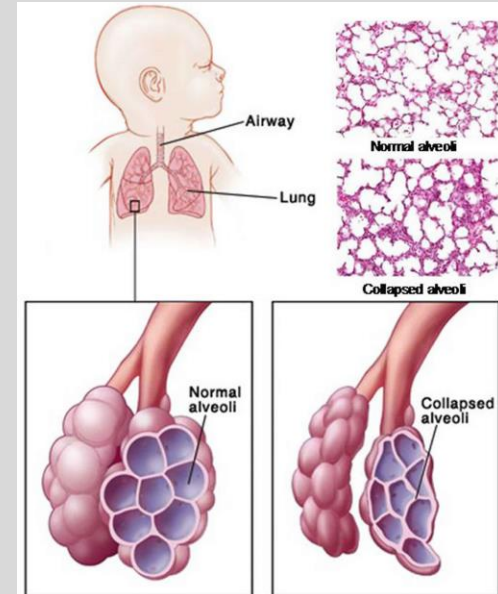
Neonatal Respiratory Distress Syndrome

Nasal continuous positive airway pressure - nCPAP

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- Born before gestational week (GW) 24:
Never surfactant
- Born GW 24-35: Uncertain



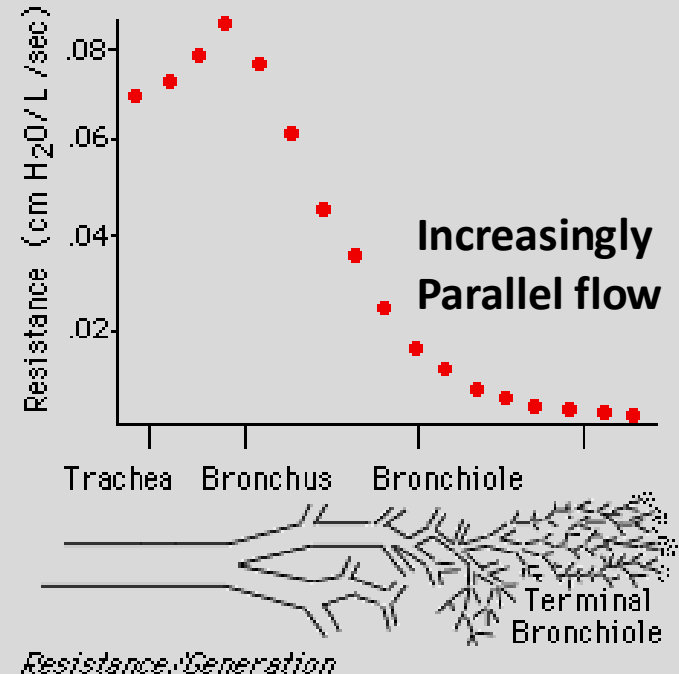
Air Flow, Pressure, and Resistance Relationships

$$Q = \frac{\Delta P}{R}$$

- Q = Air flow (L/min)
- ΔP = Pressure gradient (mm Hg or cm H₂O)
- R = Airway resistance (cm H₂O/L per second)

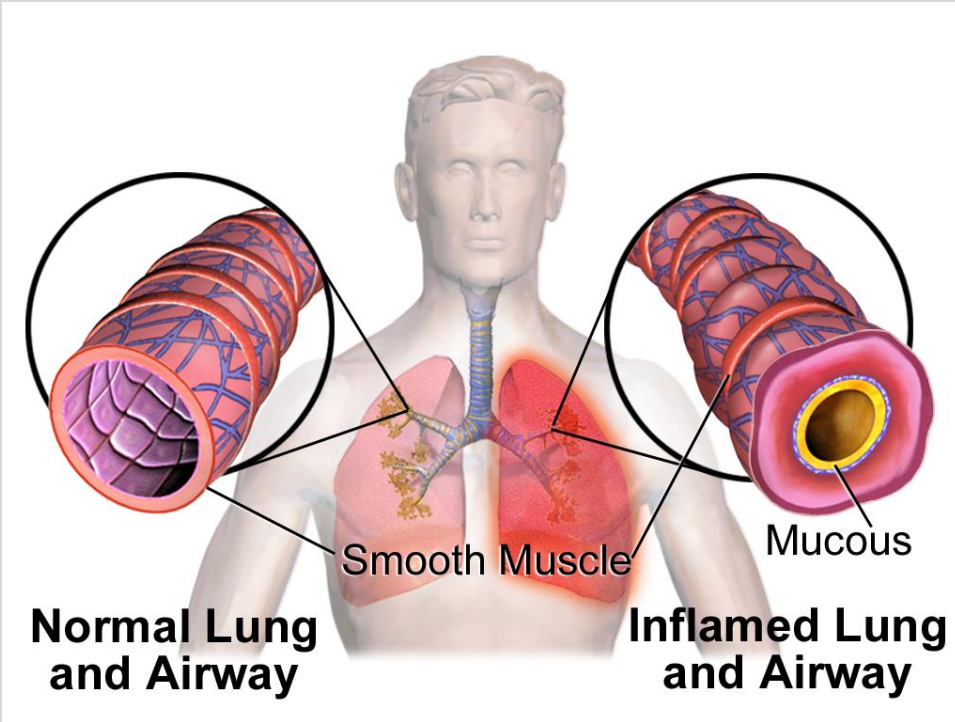
$$R = \frac{8\eta l}{\pi r^4} \quad \text{Poiseuille's Law}$$

- R = Resistance
- η = Viscosity of inspired air
- l = Length of the airway
- r = Radius of the airway



Changes in Airway Resistance

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Autonomic nervous system

- Sympathetic branch: Relaxation = Dilation

Adrenergic beta-receptor agonists

(epinephrine, isoproterenol, albuterol)

- Parasympathetic branch: Constriction

Muscarinic agonists (muscarine, carbachol)

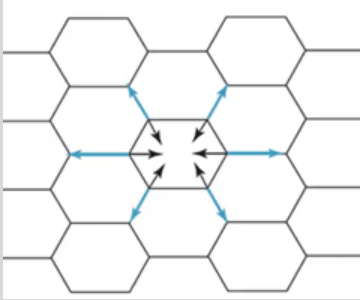
Blocked by muscarinic antagonists (atropine)

Pro-inflammatory mediators

- Histamine and leukotrienes: **Constriction**

Additional Factors Changing Airway Resistance

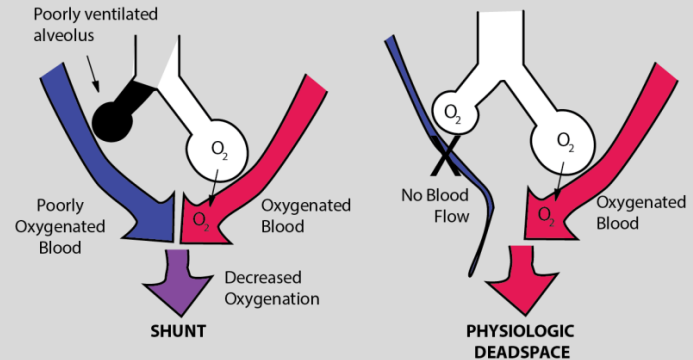
Interdependence of alveoli
e.g. in asthma



Viscosity of inspired air (η)



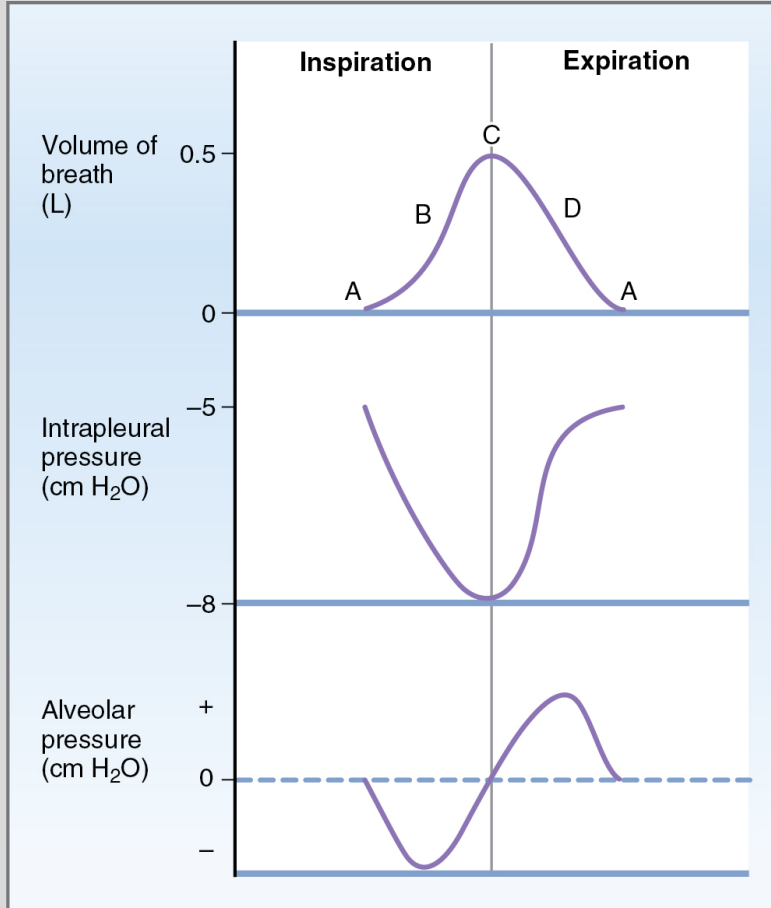
Compensatory bronchoconstriction



Alveolar PAO_2 and $PACO_2$ equal to inspired air
→ Decrease in PCO_2 → Increase in pH
→ **Bronchoconstriction**

Breathing Cycle

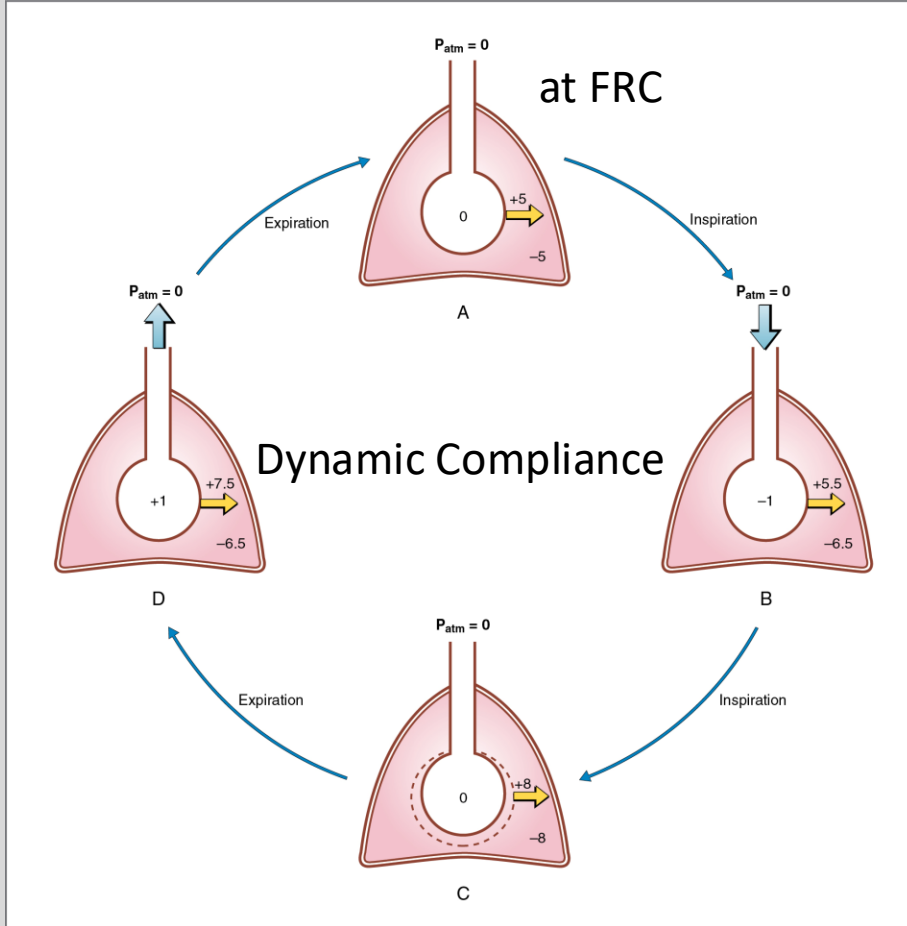
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3 Phases

Rest, Inspiration, Expiration

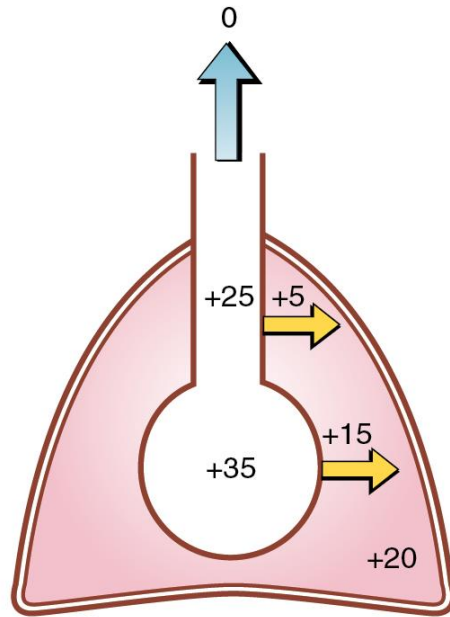
Pressures During Normal Breathing Cycle



Transmural pressure (yellow arrow) is calculated as alveolar pressure **minus** intrapleural pressure.

Pressures Across the Alveoli & Conducting Airways

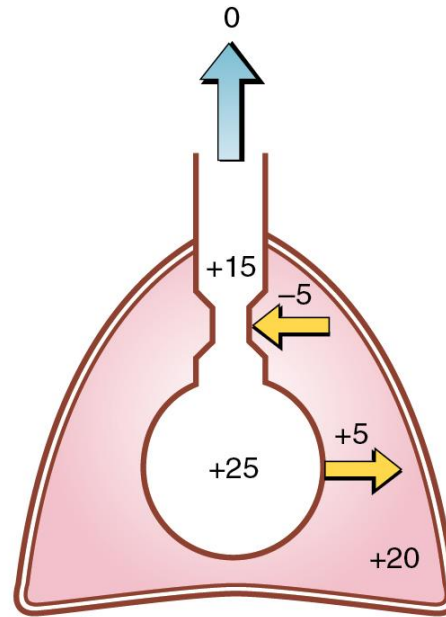
FORCED EXPIRATION



Normal

Large Airways

Alveoli



Emphysema

Transmural pressure (yellow arrow) is calculated as alveolar pressure **minus** intrapleural pressure.

Question: Will lungs collapse with positive intrapleural pressure?

Diminished elastic recoil

Medical Board Style Practice Question

A 64-year-old retired sandblaster presents with progressive shortness of breath and a dry, hacking cough. A high-resolution CT scan of the chest reveals diffuse interstitial thickening and scarring, consistent with idiopathic pulmonary fibrosis. How does his condition alter the physical forces of the respiratory system, and what is its effect on the pressure-volume loop?

- A. It increases lung compliance, shifting the pressure-volume loop upward and to the left.
- B. It decreases the inward elastic recoil of the lung, shifting the resting Functional Residual Capacity (FRC) upward.
- C. It decreases lung compliance, requiring a much larger negative intrapleural pressure to achieve a normal tidal volume.
- D. It eliminates the outward chest wall recoil, forcing the chest wall to collapse inward during quiet expiration.
- E. It decreases airway resistance, making the patient's forced expiratory volume in 1 second (FEV_1) abnormally high.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

Lecture Feedback Form

<https://www.surveymonkey.com/r/NYITCOMLectureFeedback>